Antoine Kassis

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Education

- **Masters degree MVA (Mathématiques, Vision, Apprentissage)**, ENS Paris-Saclay 2026–2027
  Geometry processing and geometric deep learning; optimal transport for machine learning; apprentissage pour les séries temporelles; robotics; probabilistic graphical models and deep generative models; foundations of LLM (not graded); stochastic calculus in machine learning; modèles de diffusion et multi-échelle en intelligence artificielle; kernel methods for machine learning; géométrie et espaces de formes; stopping times and online algorithms; training and deploying large scale models.
- **Bachelor Double Degree, Mathematics Magistère**, Université Paris-Saclay — *mention très bien* 2024–2025
- **French Baccalaureate**, Antonine International School, Lebanon — *mention très bien* 2021–2022

Research Experience

- **Research internship (TER)**, Institut Mathématique d'Orsay / INRIA DataShape — Marc Glisse, 19/20 Apr–Jun 2026
  Generalized Cohen-Steiner's curvature inequalities from curves and surfaces to arbitrary dimension. The main theorem bounds differences in Lipschitz–Killing curvatures by integrating the Euler characteristic over hyperplane sections. Persistent homology, integral geometry (Crofton, Steiner, Hadwiger), Morse theory for manifolds with boundary.
- **Research internship**, Institut Mathématique d'Orsay — Yves de Laszlo Mar–May 2025
  Co-authored a text on group representation theory from finite to solvable groups; proved foundational results including the Lie-Kolchin theorem.

Professional Experience

- **AI Automation and Platform Development**, Kite Insurance, Lebanon Jul–Sep 2026
  Helped develop 360 Broker, Kite's brokerage platform (policies, renewals, commissions, double-entry accounting), and CompareLinqs, its price comparison across 11 insurers. Built agentic AI workflows (Claude, Grok, Meta AI on WhatsApp) automating policy data entry, accounting entries, renewal follow-ups and client messages.
- **Automatants**, CentraleSupélec student AI association Nov 2025–Apr 2026
  Deep learning projects (GAN image generation, CNN for Alzheimer's detection); helped run clustering training sessions for members.
- **Product Engineering Internship (data analysis)**, TLD (ALVEST Group) Jun–Jul 2025
  Diagnosed failure causes from error logs of electric aircraft tractors. DBSCAN temporal clustering to separate root causes from secondary effects, then two composite scores — a cause/effect axis from a Beta-law fit of in-burst positions and a recurrent/impulsive axis from the dispersion of appearance days — with a rank-based metric for timestamp collisions, validated against hierarchical clustering. Delivered as an interactive Python/Dash tool for engineers.

Selected Projects (github.com/aelkas)

- **DiffusionNet on RNA surfaces** (PyTorch): implemented DiffusionNet from scratch (learned per-channel heat diffusion in the Laplace–Beltrami eigenbasis, rotation-invariant gradient features) for 160-class segmentation of RNA meshes: 84.6% validation accuracy, falling to 30% without diffusion in ablations.
- **PointNet** (PyTorch): implemented PointNet, with input and feature transform networks and an orthogonality regularizer, for classification and segmentation of 3D shapes.
- **Shaaer — Arabic Poetry Transformer** (Python): GPT trained from scratch on classical Arabic poetry with a custom grapheme-cluster-aware BPE tokenizer; ∼3.01 bits per character, below the bzip2 baseline.
- **Quad-trees** (C): binary images as quadtrees with morphological operations and a compression pass sharing structurally identical subtrees, turning the tree into a DAG.

Skills and Seminars

- **Programming:** Python, C, C++, Java, OCaml · **Databases:** PostgreSQL, Neo4j · **Tools:** LaTeX, Git
- **Languages:** French (fluent), English (TOEFL iBT 108/120), Arabic (fluent)
- **Seminars:** DataShape seminars, INRIA Saclay, on the recommendation of Nina Otter (2026); summer school on percolation under Thomas Hutchcroft, IHES (2025); Philosophy and Mathematics seminars, ENS PSL (2023–2024)
- **Interests:** Violin, Calisthenics, Swimming.